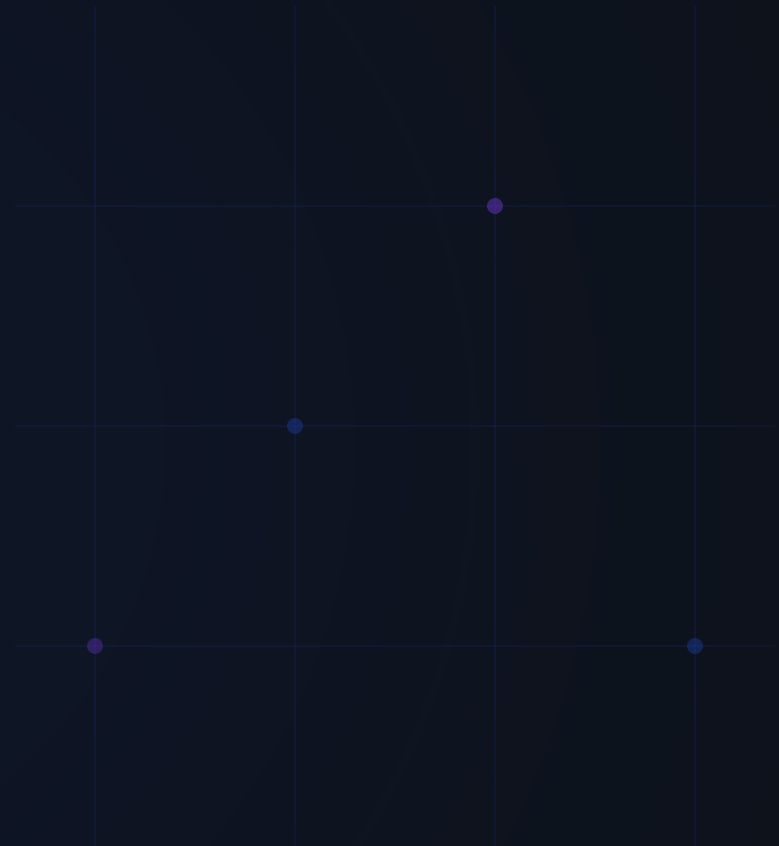




Cloudflare Radar 2024 Year in Review

A data-driven analysis of global Internet traffic, adoption trends, connectivity disruptions, and security threats

CLOUDFLARE RADAR ANNUAL REPORT



Global Internet traffic grew 17% YoY in 2024; SpaceX Starlink traffic surged 3.3x as satellite connectivity expanded to underserved regions

Global Traffic Growth

17% YoY

Starlink Traffic Surge

3.3x growth

Mobile Traffic Share

41% of global traffic

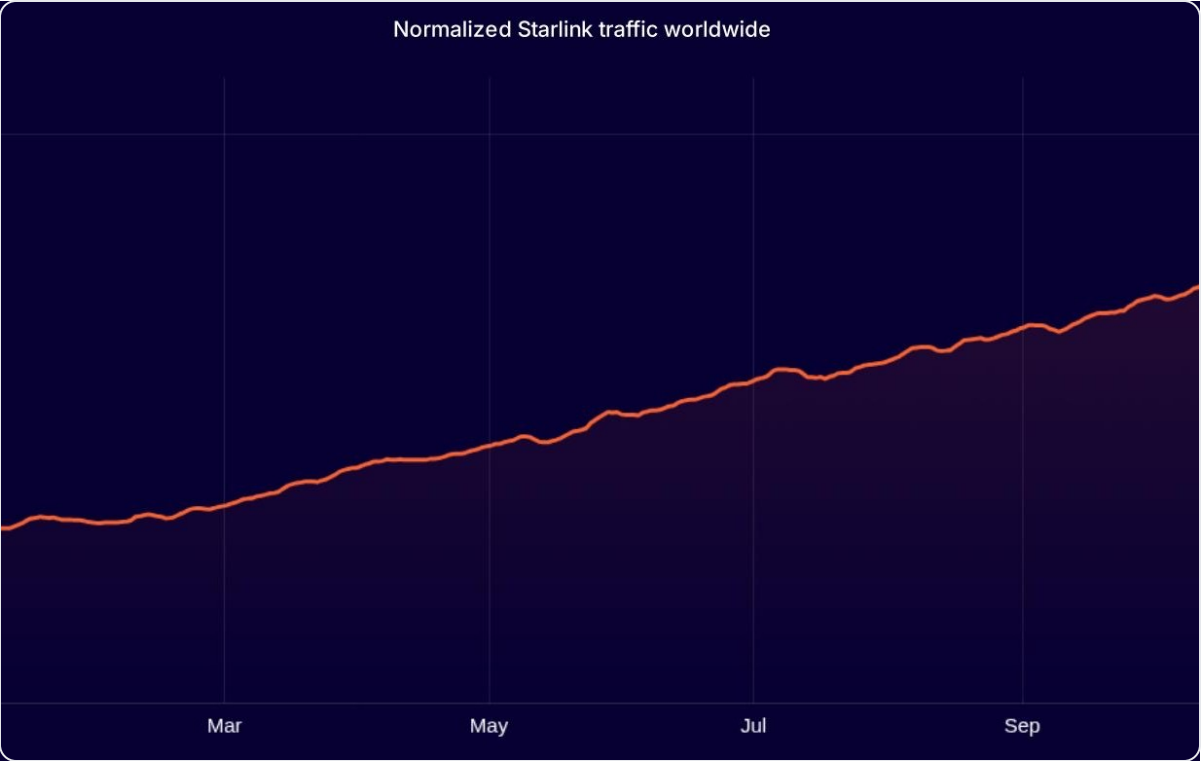
iOS Traffic Share

33% from iOS devices

Key Insights

- 17% YoY growth continues the upward trend driven by increasing reliance on the Internet
- Starlink 3.3x surge: satellite Internet expanding to previously underserved locations globally
- 41% mobile + 33% iOS reflects mobile-first connectivity patterns worldwide

Normalized Starlink traffic worldwide



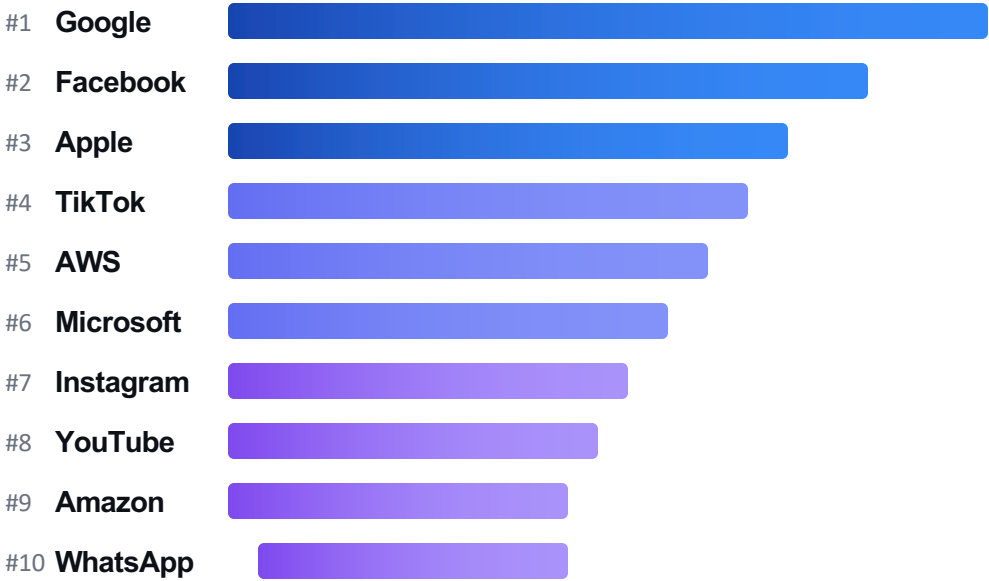
Source: Cloudflare Radar — Normalized Starlink traffic worldwide showing 3.3x growth in 2024

Search Engine Market Share



Google maintained #1 for the 2nd consecutive year; Generative AI continued its rapid rise while the Metaverse receded

Top 10 Most Popular Internet Services (2024)



Based on aggregate data from Cloudflare's 1.1.1.1 DNS resolver

Rank	Service
#1	Google
#2	Facebook
#3	Apple
#4	TikTok
#5	AWS
#6	Microsoft
#7	Instagram
#8	YouTube
#9	Amazon
#10	WhatsApp

 Generative AI: The Defining Trend of 2024

Generative AI continued its rapid rise throughout 2024, reshaping search, content creation, and developer workflows. The Metaverse receded into the background as AI-dominated services attracted the most traffic and attention. Chrome remains the most popular browser globally; Google dominates search with 89% of referral traffic.

Post-quantum encryption adoption leapt from ~2% to ~13% of TLS 1.3 traffic after Chrome 124's default enablement in April 2024



Post-Quantum Encryption (TLS 1.3)

2% → 13%

After Chrome 124 default enablement (April 17, 2024)

Browser PQC Rollout Status

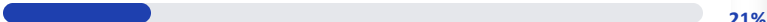
- Chrome 124: Enabled by default (April 2024)
- Firefox: Rolling out post-quantum by default
- Safari: Initial testing phase



HTTP/3 Adoption

21% of traffic

QUIC-based: faster connections,
packet-loss mitigation, encryption by default



Top API Client Language

Go ~12%

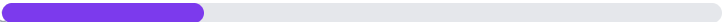
- | | | | |
|---|---------------|---|-------------|
| 1 | Go — 12% | 2 | Node.js |
| 3 | Python — 9.6% | 4 | Java — 7.4% |
| 5 | .NET — 3.6% | | |



IPv6 Adoption (Dual-Stack)

28% of traffic

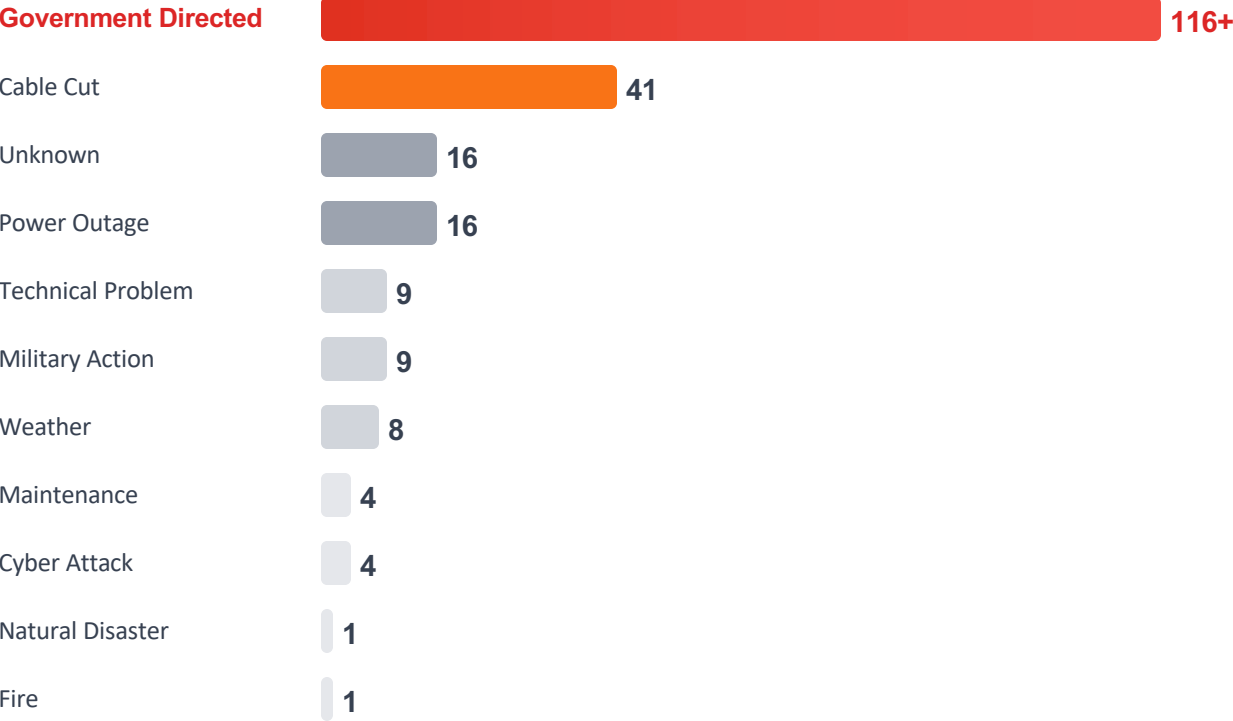
IPv6 adoption reached 28% of
dual-stack traffic during 2024



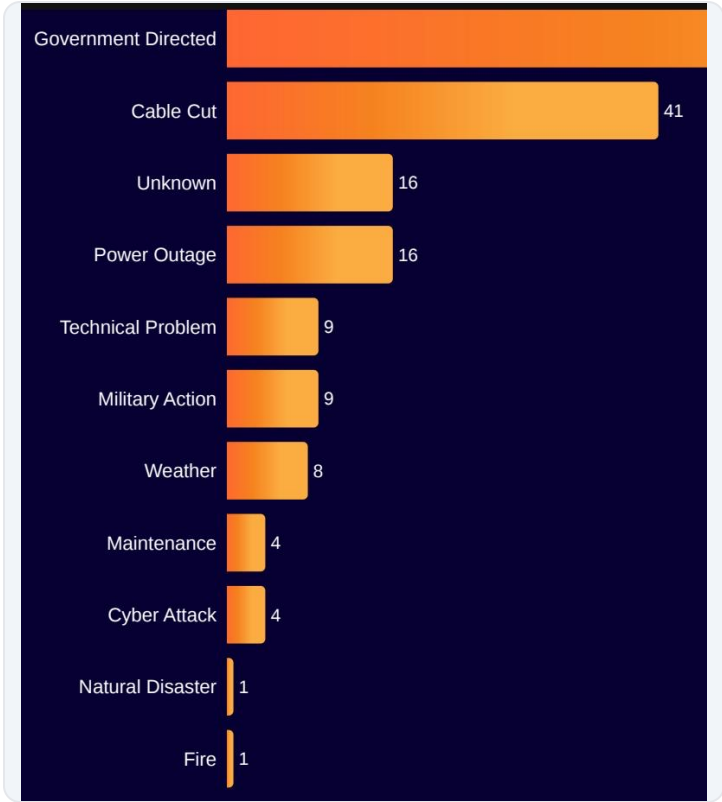
Government-directed shutdowns (116+) were the leading cause of 225 major disruptions — more than all other causes combined

Total Disruptions
225 in 2024

Internet Disruptions by Cause



Scale: 1px ≈ 3.6 disruptions | Bar width proportional to count



GOVERNMENT SHUTDOWNS

52%

of all disruptions
were government-directed
(116+ of 225 total)

More than all other
causes combined

 Geopolitical Internet Fragility

Government-directed shutdowns remain the #1 cause of Internet disruptions, reflecting a persistent trend of state-controlled access. Tracked via Cloudflare Radar Outage Center using traffic data.

6.5% of all traffic was mitigated; Gambling/Games was the most attacked sector; Log4j exploitation persists 3+ years after disclosure

Traffic Mitigated by Cloudflare

6.5% of all traffic

Most Attacked Industry

Gambling & Games

Highest attack volume among all industry categories

Bot Traffic Concentration

68% of global bot traffic from top 10 countries

RPKI Valid Routes

+6.4% increase in 2024

Log4j: Still Actively Exploited

CVE-2021-44228

Disclosed December 2021 — still actively targeted by attackers more than 2 years later

Key insight: Widely known vulnerabilities remain a critical attack surface long after initial disclosure

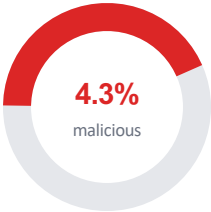
Email Security Threats

Malicious Emails

4.3%

Contain Deceptive Links

43%



Of malicious emails:

- 43% contain deceptive links
- .bar TLD: largest malicious/spam source

Security Threat Landscape 2024

Persistent threats demand continuous vigilance and layered defense

Key Security Metrics

- 6.5% of all traffic mitigated 6.5%
- Gambling/Games: most attacked sector #1
- Bot traffic from top 10 countries 68%
- Log4j still actively exploited (3+ years) Active
- Emails processed as malicious 4.3%
- Malicious emails with deceptive links 43%
- RPKI valid routes growth +6.4%

Five Takeaways Shaping the 2025 Internet Outlook

01 Traffic Growth Trajectory

17% YoY growth with Starlink's 3.3x surge signals expanding global connectivity and new infrastructure demands

Satellite Internet is bridging the digital divide at unprecedented speed

02 Post-Quantum Encryption Momentum

Jump from 2% to 13% of TLS 1.3 traffic demands crypto-agility in 2025 security roadmaps

Chrome, Firefox, and Safari are all moving toward PQC by default

03 Connectivity Fragility

Government-directed shutdowns (116+) expose geopolitical Internet risk — more than all other disruption causes combined

Organizations need resilient, multi-path connectivity strategies

04 Persistent Security Threats

6.5% traffic mitigated, Log4j exploitation continues — legacy vulnerabilities remain a critical attack surface

Patch management and continuous monitoring are non-negotiable

05 Evolving Developer Ecosystem

Go's rise as the top API client language (~12%) reflects infrastructure automation and cloud-native development trends. HTTP/3 adoption at 21% signals the protocol transition is accelerating.

API-first architectures are reshaping how services communicate and scale

The Internet in 2025 will be faster, more encrypted, and more contested.

Strategic planning must account for quantum readiness, geopolitical resilience, and evolving threat landscapes.

CLOUDFLARE RADAR 2024